THE LISSE-ETALE SITE IS NOT FUNCTORIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Standard warning example for the lisse-etale topology on algebraic stacks.

Editorial note. The label lemma-lisse-etale-not-functorial names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The lisse-etale site of a scheme X: the category of schemes smooth over X with the usual etale coverings (Cohomology of Stacks, Section stacks-cohomology-section-lisse-etale)
- A continuous functor u between the underlying categories induces an adjoint pair of functors u-s and u-small on sheaf categories (Sites, Section sites-section-continuous-functors)
- Sites, Definition sites-definition-morphism-of-sites: u defines a morphism of sites exactly when the pushforward u-small is left exact
- Sites, Lemma sites-lemma-pullback-representable-sheaf: the pushforward of a representable sheaf under base change is the representable sheaf of the fibre product

Source reference. Original definition 00X1: sources/stacks-chow/tags.txt, source lines 1203–1203. Prerequisite pointer invoked at 4822; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 04D3: sources/stacks-chow/tags.txt, source lines 5713–5713. Prerequisite pointer invoked at 4825; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 00WU: sources/stacks-chow/tags.txt, source lines 1196–1196. Prerequisite pointer invoked at 4820; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0786: sources/stacks-chow/tags.txt, source lines 9226–9226. Prerequisite pointer invoked at 4801; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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The *lisse-étale* site $X_{lisse,étale}$ of X is the category of schemes smooth over X endowed with (usual) étale coverings, see Cohomology of Stacks, Section 0786. Let $f : X \to Y$ be a morphism of schemes. There is a functor

$$u : Y_{lisse,étale} \longrightarrow X_{lisse,étale}, \quad V/Y \longmapsto V \times_Y X$$

which is continuous. Hence we obtain an adjoint pair of functors

$$u^s : Sh(X_{lisse,étale}) \longrightarrow Sh(Y_{lisse,étale}), \quad u_s : Sh(Y_{lisse,étale}) \longrightarrow Sh(X_{lisse,étale}),$$

see Sites, Section 00WU. We claim that, in general, u does **not** define a morphism of sites, see Sites, Definition 00X1. In other words, we claim that u_s is not left exact in general. Note that representable presheaves are sheaves on lisse-étale sites. Hence, by Sites, Lemma 04D3 we see that $u_s h_V = h_{V \times_Y X}$. Now consider two morphisms

$$\begin{array}{ccc} V_1 & \overset{a}{\underset{b}{\rightrightarrows}} & V_2 \\ & \searrow \quad \swarrow & \\ & Y & \end{array}$$

of schemes V_1, V_2 smooth over Y. Now if u_s is left exact, then we would have

$$u_s \text{Equalizer}(h_a, h_b : h_{V_1} \to h_{V_2}) = \text{Equalizer}(h_{a \times 1}, h_{b \times 1} : h_{V_1 \times_Y X} \to h_{V_2 \times_Y X})$$

We will take the morphisms $a, b : V_1 \to V_2$ such that there exists no morphism from a scheme smooth over Y into $(a = b) \subset V_1$, i.e., such that the left hand side is the empty sheaf, but such that after base change to X the equalizer is nonempty and smooth over X. A silly example is to take $X = \text{Spec}(\mathbf{F}_p)$, $Y = \text{Spec}(\mathbf{Z})$ and $V_1 = V_2 = \mathbf{A}^1_{\mathbf{Z}}$ with morphisms $a(x) = x$ and $b(x) = x + p$. Note that the equalizer of a and b is the fibre of $\mathbf{A}^1_{\mathbf{Z}}$ over (p).

Lemma 0.1. *The lisse-étale site is not functorial, even for morphisms of schemes.*

Proof. See discussion above. □